Naïve Bayes (NB) classifier using the Iris dataset

Python codes for Naïve-Bayes (NB) classifier using the Iris dataset and then testing it with an unknown sample (Jupyter Notebook)

1. Import Libraries

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, classification_report
```

2. Load the Iris Dataset

In [2]:
```
iris = load_iris()

X = iris.data
y = iris.target

print("Feature names:", iris.feature_names)
print("Target names:", iris.target_names)
```

```
Feature names: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal
width (cm)']
Target names: ['setosa' 'versicolor' 'virginica']
```

3. Split the Dataset

In [3]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training samples:", X_train.shape[0])
print("Testing samples:", X_test.shape[0])
```

```
Training samples: 120
Testing samples: 30
```

4. Train Naïve-Bayes Classifier

In [4]:
```
nb = GaussianNB()
nb.fit(X_train, y_train)
```

Out[4]:
```
GaussianNB()
```

5. Evaluate the Model

In [5]:
```
y_pred = nb.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n",
      classification_report(y_test, y_pred, target_names=iris.target_names))
```

```
Accuracy: 1.0

Classification Report:
               precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      1.00      1.00         9
   virginica       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

6. Predict Unknown Sample

In [6]:
```
# Unknown flower (sepal length, sepal width, petal length, petal width)
unknown_sample = np.array([[5.1, 3.5, 1.4, 0.2]])

prediction = nb.predict(unknown_sample)
predicted_class = iris.target_names[prediction[0]]

print("Predicted class for unknown sample:", predicted_class)
```

```
Predicted class for unknown sample: setosa
```

7. Test Multiple Unknown Samples

In [7]:
```
unknown_samples = np.array([
    [6.7, 3.1, 4.7, 1.5],
    [7.2, 3.6, 6.1, 2.5],
    [5.0, 3.4, 1.5, 0.2]
])

predictions = nb.predict(unknown_samples)

for i, pred in enumerate(predictions):
    print(f"Sample {i+1} → {iris.target_names[pred]}")
```

```
Sample 1 → versicolor
Sample 2 → virginica
Sample 3 → setosa
```